

Week 4 Decision Gate: *King Octopus and the servants*

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Sources

The definitions I rely on (implication; “always lies”/“always tells the truth”; divisibility; ...) and the exact problem statement I chose.

Rules / Assumptions

What the key terms mean here: *valid*, *follows*, “for every integer”, a servant has exactly one of 6, 7, 8 legs, ...

Work

Claim. *State the claim precisely.*

Proof. Steps, each justified by a definition or a prior step. Or, if the claim is false: □

Counterexample. *An object in the claim’s domain that satisfies the hypothesis and violates the conclusion. Show the values.*

Check Your Answer

Either verify *every* proof step against a definition, or verify the counterexample really satisfies the hypothesis and violates the conclusion. For an argument-validity problem, exhibit premises that are all true while the conclusion is false (or argue that is impossible).

One-Sentence Summary

Exactly what you established — and, where it matters, distinguish “the conclusion happens to be true” from “the argument is valid.”

Strongest disagreement

The best objection a sharp, skeptical reader would raise, stated honestly, and whether it survives.